Shashank Mishra had never heard of chronic obstructive pulmonary disease (COPD) till his 69-year-old father was diagnosed with it last December. Since then, life has turned upside down for the Delhi-based engineer, because "there is nothing more difficult than watching a parent struggle for breath". Common COPD symptoms include breathlessness, constant mucus formation and wheezing, which prevents many patients from falling asleep. "My father is now going to respiratory rehab, but since there is only one centre near our house, getting an appointment takes long," says Mishra.

It is a problem many in the country have found themselves facing in the past few years. "COPD is not Covid or asthma," says Dr Vivek Singh, director of respiratory and sleep medicine at the Medanta hospital in Gurugram. "It is an irreversible condition where one's airways get blocked and a patient can't breathe. In severe cases, even taking two steps becomes difficult." And while numbers are hard to come by, cases of COPD have been increasing steadily over the years.

According to the global burden of disease (GBD), a tool that helps measure mortality across countries, time, age and sex, COPD is the second leading cause of death and disability-adjusted life years (DALYs) in India. Based on prevalence studies, mortality data and the burden of risk factors for COPD across India, the 2019 data for GBD estimated COPD prevalence to be 37.8 million cases accounting for 9.5 per cent of the deaths in India, from 28.1 million cases in 1990. "It is a growing burden in India," says Dr Randeep Guleria, pulmonologist and former director of AIIMS, Delhi. "While smoking was a big risk factor previously for developing COPD later on in life, air pollution is playing a major role now."

The 2019 GBD numbers bear this out. COPD caused by non-smoking factors accounted for 48 per cent of COPD deaths and 51 per cent of COPD DALYs in India. "COPD develops as a result of exposure to inhaled particles, which irritate the airways," says Dr Sudarshan K.S., pulmonologist at the Fortis hospital in Bengaluru. "When you are exposed to such particles for a long period of time, they cause inflammation in the airways, making them narrow and causing breathing difficulties. In some patients, the irritants can also damage the lung alveoli, where oxygen and carbon dioxide exchange happens." Pulmonary tuberculosis and poorly-treated asthma are other risk factors associated with COPD, both of which are highly prevalent in India.

With the past three years consumed by Covid-19 and knowledge about its impact, whether short- or long-term, still limited, doctors are not sure if exposure to the virus directly causes COPD or aggravates an existing condition. "We have seen patients having COPD-type obstruction in their airways after Covid, but it is too early to say whether this is a permanent condition or not," says Dr Sandeep Nayar, senior director and the head of the department of chest and respiratory diseases at the BLK-Max Super Speciality Hospital in Delhi. "But Covid being a predominantly lung condition, it follows that it will contribute to the worsening of risk factors for COPD."

THE IMPACT OF TOXIC AIR

There is mounting clinical and theoretical evidence of direct correlation between air pollution and COPD risk. A 15-year study of 5.1 million adults in *ATS Journals* in 2019 found air pollution to lead to a higher incidence of COPD. "Air pollution can damage multiple organs, but the lungs are a main target," says Dr Rajesh Chawla, pulmonologist at the Apollo Hospitals in Delhi. "This is because the respiratory tract has direct exposure to it."

The chief villains of air pollution—particulate matter, sulphur dioxide, nitrogen dioxide, carbon monoxide and lead—can inflame the respiratory tract. For those already suffering from COPD, rising pollution means more visits to the emergency room. A 1992 research in online resource PubMed showed the increased risk of dying of COPD associated with air particles among the general population, thrice that of all-cause mortality. This, doctors observe, holds true even today. "The period when pollution is high in Delhi, I know I'll be in hospital with an oxygen mask for a week or so," says Delhi-based Kalpana Mitra, a retired banker who was diagnosed with COPD five years ago at age 65. "This has happened three years in a row. My children want me to leave the city, but at this age, moving is hard. The solution is to sit in a room with an air filter on for 24 hours till pollution levels reduce." Doctors also advise strongly against exercise and overexposure to polluted air as the AQI (air quality index) in major cities hovers at toxic levels. "Early mornings and evenings are bad times to exercise outdoors when pollution is high," says Dr Prashant Chhajed, head of pulmonology at the Fortis hospital in Vashi, Mumbai. "I would also recommend using a good-quality mask and air filters."



"WHILE SMOKING WAS A BIG RISK FACTOR EARLIER FOR DEVELOPING COPD, AIR POLLUTION IS PLAYING A MAJOR ROLE NOW"

DR RANDEEP GULERIA
Pulmonologist, former Director, AIIMS